

Phan Van Manh

Full-Stack Software Engineer

LinkedIn: <https://www.linkedin.com/in/phan-manh-34a107269>

Website: <https://ducmanh001.github.io/manh-fullstack-portfolio>

Mobile: 0845 012 095

Email: manhphan110801@gmail.com

GitHub: <https://github.com/ducmanh001>

Location: Ha Noi

SUMMARY

Full-Stack Software Engineer with over 3 years of experience building NestJS APIs and React/Next.js interfaces. Built a real-time voice-call app and admin console around a shared OpenAPI-generated client, redesigned an analytics pipeline over 300M+ survey records for consistent 1-2s aggregations, and cut a clinic-classification SQL query from ~600ms to ~40ms.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript

Frontend: React, Next.js, React Router, Vite; TanStack Query, Zustand, React Hook Form, Zod; Ant Design, shadcn/ui, Tailwind CSS; Socket.IO, WebRTC (LiveKit)

Backend: Node.js, NestJS

Data: PostgreSQL, ClickHouse, OpenSearch, MongoDB, Redis; Prisma, TypeORM

Messaging & APIs: BullMQ, Kafka; REST, gRPC, WebSocket

Testing: Jest, Supertest, Cypress, Playwright

Delivery: Docker, CI/CD on DigitalOcean

Process: Agile/Scrum, TDD, AI-Driven Development (AI-DD)

Principles: OOP, SOLID, DRY, KISS, YAGNI, Design Patterns

Architecture: Clean/Layered Architecture, Modular Monolith, Microservices, DDD, CQRS, Event-driven Design

EXPERIENCE

Sun Asterisk

Jul 2025 - Present

Backend Software Engineer

- Proposed and led a rebuild of the respondent-analytics module for a Japanese client: replaced OpenSearch parent-child aggregation queries on a single unpartitioned index with a ClickHouse columnar table partitioned by year, validated against a 300M+ row production copy spanning 50+ years of survey data, where it eliminated recurring query timeouts and cut typical aggregation latency to a consistent 1-2s.
- Adapted an internal AI-assisted delivery workflow (AI-DD) for a Japanese client, adding source-commit traceability so AI-generated design docs are verified against their origin before reaching developers, catching unsupported content before downstream rework.

Fenik Technologies

May 2023 - Jul 2025

Full-Stack Software Engineer

- Developed and maintained features for a dental-clinic discovery and review platform serving 100,000+ monthly visits.
- Designed and implemented a high-performance full-text search API using PostgreSQL's native full-text search (tsvector + GIN index) with custom relevance ranking, avoiding the cost and operational overhead of a dedicated search engine for the platform's data scale, cutting search response times by over 10x.
- Refactored the clinic classification API by partitioning the table by region and optimizing SQL queries, reducing query execution time from ~600ms to ~40ms.

PERSONAL PROJECTS

Litmatch

2024 - Present

GitHub: <https://github.com/ducmanh001/litmatch> | Demo: <https://litmatch-web.manhphan110801.workers.dev>

- Solo-designed and built Litmatch - an anonymous social-entertainment platform (matching, voice/text chat, feed, mini-games, diamond/VIP economy) - as a Next.js web client, a Vite/React admin console, and a NestJS modular-monolith core API across 20+ modules, both frontends sharing a single typed REST client generated from the OpenAPI spec.
- Built the voice-match call flow end-to-end: REST join mints a LiveKit token per match session, a React hook connects the room client-side guarded by a generation counter against stale reconnects, and Socket.IO delivers real-time call-ended events with REST poll as fallback.

Mini Shop API

2025 - Present

GitHub: <https://github.com/ducmanh001/mini-shop-api> | Demo: <https://mini-shop-api-production-e25a.up.railway.app/api-docs>

- Solo-built a NestJS/TypeScript e-commerce backend (auth, products, cart, checkout, orders, reviews, admin) with PostgreSQL/TypeORM migrations and Bull-queued email notifications; transactional checkout uses ordered row locks and idempotency keys to prevent duplicate orders. Deployed to Railway with Docker and CI.

EDUCATION

FPT Polytechnic

Sep 2020 - Feb 2023

Web Programming